CS-300 Analysis and Design

Project One: Pseudocode for ABC University

Southern New Hampshire University

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PSEUDOCODE for ABC University

```
// Loading the file, reading the lines, parsing, and checking format
START PROGRAM
CREATE Vector<Course> courseList
OPEN the course file
IF the file does not open
        PRINT “File Error”
        STOP PROGRAM
END IF

FOR each line in the file
        SPLIT the line by commas into tokens

        IF tokens count < 2
                PRINT “Line format error”
                CONTINUE
        END IF

        SET courseNumber = first token
        SET courseTitle = second token

        CREATE empty Vector<String> prereqList

        FOR each token after the second token
                ADD token to prereqList
        END FOR
```

```
        CREATE Course tempCourse
        tempCourse.number = courseNumber
        tempCourse.title = courseTitle
        tempCourse.prerequisites = prereqList

        ADD tempCourse to courseList
END FOR
CLOSE file

// check that the prerequisites exist
FOR each course in courseList
        FOR each prereq in course.prerequisites
                SET found = FALSE

                FOR each course2 in courseList
                        IF course2.number == prereq
                                found = TRUE
                        END IF
                END FOR

                IF found == FALSE
                        PRINT "Prerequisite " + prereq + " not found"
                END IF
        END FOR
END FOR

//Course object structure and storing the vector (loop)
```

```
STRUCT Course
        STRING number
        STRING title
        Vector<String> prerequisites
END STRUCT

// Searching for a course and printing it
FUNCTION searchCourse(courseList, courseNumber)
        FOR each course in courseList
                IF course.number == courseNumber
                        PRINT course.number + ": " + course.title

                        IF course.prerequisites is empty
                                PRINT "No prerequisites"
                        ELSE
                                PRINT "Prerequisites:"
                                FOR each prereq in course.prerequisites
                                        PRINT prereq
                                END FOR
                        END IF

                        RETURN
                END IF
        END FOR

        PRINT "Course not found"
END FUNCTION
```